

CS51 - Assignment 2

Are you an introvert?

Due: Sep 11 at 11:59pm

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1 Part 1: Disjunctive and Conjunctive Normal Forms

A	B	C	D	$minterm$	$maxterm$	I
0	0	0	0	$m_0 = \neg A \wedge \neg B \wedge \neg C \wedge \neg D$	$M_0 = A \vee B \vee C \vee D$	0
0	0	0	1	$m_1 = \neg A \wedge \neg B \wedge \neg C \wedge D$	$M_1 = A \vee B \vee C \vee \neg D$	0
0	0	1	0	$m_2 = \neg A \wedge \neg B \wedge C \wedge \neg D$	$M_2 = A \vee B \vee \neg C \vee D$	0
0	0	1	1	$m_3 = \neg A \wedge \neg B \wedge C \wedge D$	$M_3 = A \vee B \vee \neg C \vee \neg D$	1
0	1	0	0	$m_4 = \neg A \wedge B \wedge \neg C \wedge \neg D$	$M_4 = A \vee \neg B \vee C \vee D$	0
0	1	0	1	$m_5 = \neg A \wedge B \wedge \neg C \wedge D$	$M_5 = A \vee \neg B \vee \neg C \vee \neg D$	0
0	1	1	0	$m_6 = \neg A \wedge B \wedge C \wedge \neg D$	$M_6 = A \vee \neg B \vee \neg C \vee D$	1
0	1	1	1	$m_7 = \neg A \wedge B \wedge C \wedge D$	$M_7 = A \vee \neg B \vee \neg C \vee \neg D$	1
1	0	0	0	$m_8 = A \wedge \neg B \wedge \neg C \wedge \neg D$	$M_8 = \neg A \vee B \vee C \vee D$	0
1	0	0	1	$m_9 = A \wedge \neg B \wedge \neg C \wedge D$	$\neg M_9 = \neg A \vee B \vee C \vee \neg D$	0
1	0	1	0	$m_{10} = A \wedge \neg B \wedge C \wedge \neg D$	$M_{10} = \neg A \vee B \vee \neg C \vee D$	1
1	0	1	1	$m_{11} = A \wedge \neg B \wedge C \wedge D$	$M_{11} = \neg A \vee B \vee \neg C \vee \neg D$	1
1	1	0	0	$m_{12} = A \wedge B \wedge \neg C \wedge \neg D$	$M_{12} = \neg A \vee \neg B \vee C \vee D$	1
1	1	0	1	$m_{13} = A \wedge B \wedge \neg C \wedge D$	$M_{13} = \neg A \vee \neg B \vee C \vee \neg D$	1
1	1	1	0	$m_{14} = A \wedge B \wedge C \wedge \neg D$	$M_{14} = \neg A \vee \neg B \vee \neg C \vee D$	1
1	1	1	1	$m_{15} = A \wedge B \wedge C \wedge D$	$M_{15} = \neg A \vee \neg B \vee \neg C \vee \neg D$	1

Disjunctive Normal Form: $I = m_3 \vee m_6 \vee m_7 \vee m_{10} \vee m_{11} \vee m_{12} \vee m_{13} \vee m_{14} \vee m_{15} = (\neg A \wedge \neg B \wedge C \wedge D) \vee (\neg A \wedge B \wedge C \wedge \neg D) \vee (A \wedge \neg B \wedge C \wedge \neg D) \vee (A \wedge \neg B \wedge C \wedge D) \vee (A \wedge \neg B \wedge C \wedge D) \vee (A \wedge B \wedge \neg C \wedge \neg D) \vee (A \wedge B \wedge \neg C \wedge D) \vee (A \wedge B \wedge C \wedge \neg D) \vee (A \wedge B \wedge C \wedge D) \vee (A \wedge B \wedge C \wedge D)$

Conjunctive Normal Form: $I = m_1 \wedge m_2 \wedge m_3 \wedge m_4 \wedge m_5 \wedge m_6 \wedge m_7 \wedge m_8 \wedge m_9 = (A \vee B \vee C \vee D) \wedge (A \vee B \vee C \vee \neg D) \wedge (A \vee B \vee \neg C \vee D) \wedge (A \vee B \vee \neg C \vee \neg D) \wedge (A \vee \neg B \vee C \vee D) \wedge (A \vee \neg B \vee C \vee \neg D) \wedge (A \vee \neg B \vee \neg C \vee D) \wedge (A \vee \neg B \vee \neg C \vee \neg D)$

2 Part 2: K-Maps

A	B	C	D	I	$minterm$
0	0	0	0	0	m_0
0	0	0	1	0	m_1
0	0	1	0	0	m_2
0	0	1	1	1	m_3
0	1	0	0	0	m_4
0	1	0	1	0	m_5
0	1	1	0	1	m_6
0	1	1	1	1	m_7
1	0	0	0	0	m_8
1	0	0	1	0	m_9
1	0	1	0	1	m_{10}
1	0	1	1	1	m_{11}
1	1	0	0	1	m_{12}
1	1	0	1	1	m_{13}
1	1	1	0	1	m_{14}
1	1	1	1	1	m_{15}

		CD			
		00	01	11	10
AB	00	0	0	1	0
	01	0	0	1	1
	11	1	1	1	1
	10	0	0	1	1

2.1 Boolean Conclusions from Karnaugh map

Function 1: Green

$C = 1, D = 1$

$(C \wedge D)$

Function 2: Red

$A = 1, B = 1$

$(A \wedge B)$

Function 3: Yellow:

$A = 1, C = 1$

we get

$(A \wedge C)$

Function 4: Blue

$B=1, C=1 (B \wedge C)$

Together we get the boolean function $(C \wedge D) \vee (A \wedge B) \vee (A \wedge C) \vee (B \wedge C)$

This can be confirmed visually in several ways. For example, there are no horizontal bars or which contains a one, nor is there a vertical one which contains repeating variables that evaluate to true.

This would explain why every variable has a not in front of it.

3 Extra credit

		CD			
		00	01	11	10
AB	00	0	0	1	0
	01	0	0	1	1
	11	1	1	1	1
	10	0	0	1	1

red: $A = 0, C = 0$

yellow: $AB = 0 CD = 10$

green: $AB = 10 C = 0$

$$(A \vee C) \wedge (A \vee B \vee \neg C \vee D) \wedge (\neg A \vee B \vee C)$$